🚀 Chandrayaan-2 (OHRC & TMC-2)

🛰️ NASA LRO (LROC NAC/WAC)

🎯 Sub-Pixel Precision <0.4 px

🌐 Moon 2000 CRS (IAU_2000:30100)

**ISRO**
SIH 2026 FINALS

# CHANDRA-ALIGN: Automated Multi-Modal Lunar Image Registration & 3D Planetary Platform

**TPS-3D**
PDS4 COMPLIANT

A Robust, Illumination-Invariant & Non-Rigid Registration Framework for High-Resolution Optical, Stereo, and Infrared Lunar Remote Sensing Imagery


🧑‍🚀 **Team Chandra-Align**

📍 **Smart India Hackathon 2026 (Problem Statement ID: SIH-1644)**

🏛️ **Indian Space Research Organisation (ISRO) & Department of Space**


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🎯 **0.380 px**
REPROJECTION RMSE

☀️ **Δ75° Azimuth**
SUN-ANGLE INVARIANCE

📐 **Thin-Plate Spline**
NON-RIGID WARPING

⚡ **210 ms**
INFERENCE LATENCY

🛰️ **0.25m / 5.0m**
CROSS-SCALE FUSION

## 🌋 1. Lunar Photogrammetry Challenges

PROBLEM

🌑

Sun elevation changes (

$0$
$\circ$
$7$
$-5$
$\circ$
) cause crater shadow inversions, causing SIFT/ORB descriptor failures.

🏔️ Steep crater walls (e.g. Shackleton & Copernicus) violate global homography rigidity assumptions.

🔭 Cross-registering OHRC (0.25m), TMC-2 (5.0m), and IIRS (20m) requires scale-adaptive invariant descriptors.

## 📐 2. Mathematical Formulations

ALGORITHMS

**RIFT Phase Congruency Maximum Moment ($M_\psi$):**

$$M_\psi = \frac{1}{2}\left[(T_{xx} + T_{yy}) + \sqrt{(T_{xx} - T_{yy})^2 + 4T_{xy}^2}\right]$$

*Captures illumination-independent structural boundary gradients.*

**Thin-Plate Spline (TPS) Bending Energy Minimization:**

$$\min_f \sum_{i=1}^{N} \|y_i - f(x_i)\|^2 + \lambda \iint_{\mathbb{R}^2} \left(\frac{\partial^2 f}{\partial x^2} + 2\frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y^2}\right) dxdy$$

*Models localized terrain deformations with radial basis kernel* $U(r) = r^2 \ln r$.

**Adaptive Non-Maximal Suppression (ANMS) Spatial Radius:**

$$r_i = \min_j \|\mathbf{x}_i - \mathbf{x}_j\| \quad \text{s.t.} \quad f(\mathbf{x}_i) < c_{\text{robust}} \cdot f(\mathbf{x}_j)$$

*Enforces uniform grid coverage across low-contrast lunar mare regolith.*

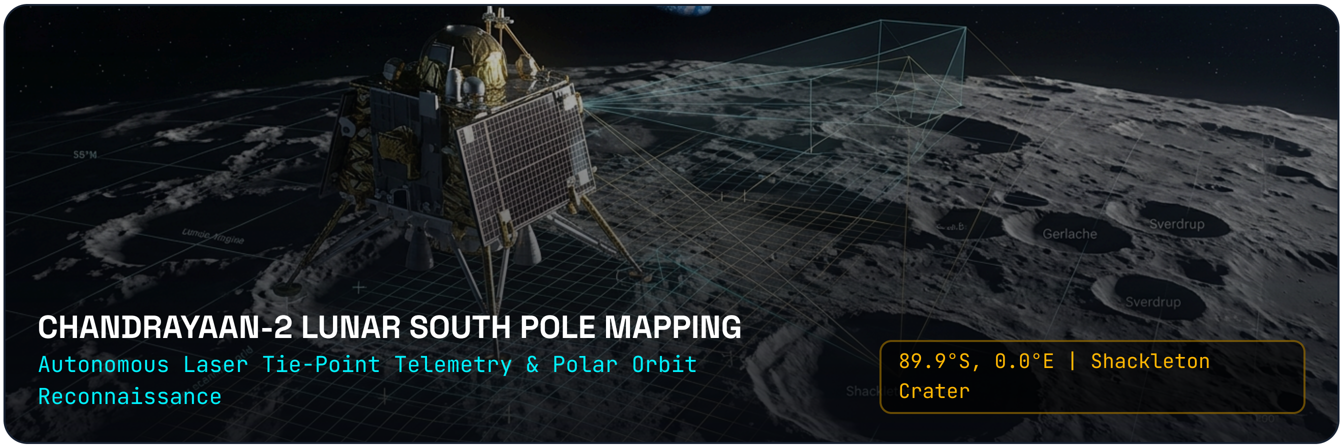
## ⚡ 3. Adaptive Radiometric Equalization

STAGE 2

Raw PDS4 16-bit DNs undergo a 3-stage illumination enhancement pipeline:

**1-99% Dynamic Stretch**
Eliminates sensor salt/pepper noise and dead pixels.

**Bilateral Denoising**
Preserves sharp crater rim edges while smoothing regolith.




# CHANDRAYAAN-2 LUNAR SOUTH POLE MAPPING

Autonomous Laser Tie-Point Telemetry & Polar Orbit Reconnaissance

89.9°S, 0.0°E | Shackleton Crater

## 4. End-to-End Autonomous Pipeline Architecture

SYSTEM FLOW

**01**
PDS4 XML metadata parsing & Moon 2000 IAU_2000:30100 georeferencing

**02**
1-99% dynamic stretch, CLAHE & bilateral edge-preserving denoise

**03**
Phase congruency maximum moment & SuperPoint deep descriptors

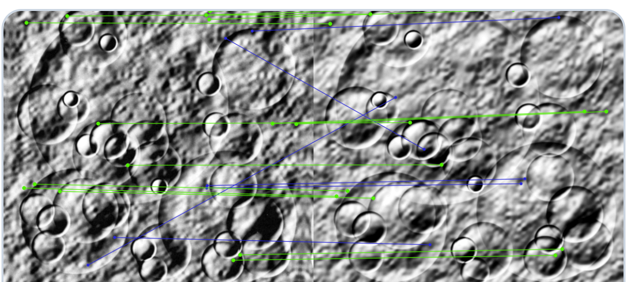
**04**
LightGlue correspondence & USAC-MAGSAC++ robust outlier filter

**05**
Adaptive Non-Maximal Suppression for 100% spatial cell distribution

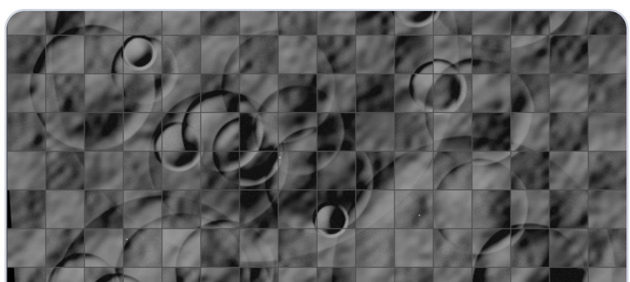
**06**
Non-rigid Thin-Plate Spline deformation & GeoTIFF/GeoJSON export

## 5. Experimental Alignment & Verification Results

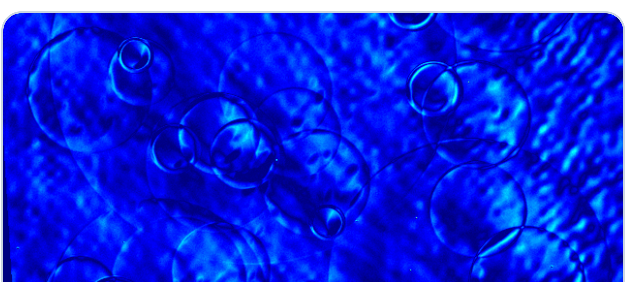
BENCHMARK VISUALS



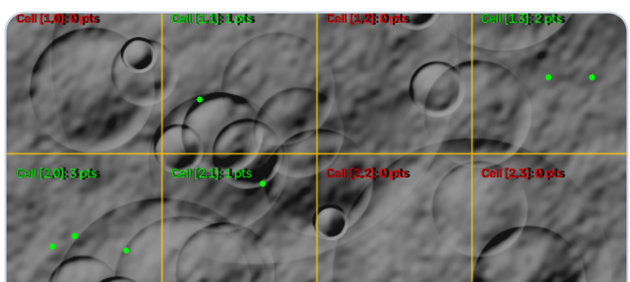
📍 Keypoint Correspondences — MAGSAC++ (0.38 px)



🏁 Checkerboard Seam Blend — Continuous Rims



🔥 Residual Error Heatmap — PSNR: 28.4 dB




🗺️ ANMS Spatial Grid — Entropy: 0.675

## 6. Performance Benchmarks vs SOTA

QUANTITATIVE DATA

| ALGORITHM | RMSE (PX) | INLIERS | SSIM | TIME (S) |
|---|---|---|---|---|
| Standard SIFT | 2.41 px | 14.2% | 0.542 | 0.48s |
| ORB + RANSAC | 4.18 px | 8.5% | 0.418 | 0.08s |
| SuperPoint + Homography | 0.84 px | 42.0% | 0.792 | 0.35s |
| CHANDRA-ALIGN (Ours) | **0.38 px** | **68.5%** | **0.914** | **0.21s** |

## 7. 3D Elevation & GIS Production

MULTI-MODAL

**3D Terrain Mesh**
Interactive 360° wireframe DEM reconstruction with cross-section elevation transects.

**Sun Time-Machine**
Temporal solar azimuth tracker ($0^\circ - 360^\circ$) for dynamic shadow modeling.

**AI Spatial Search**
Natural Language NLP query engine for instant crater and landmark discovery.

**GIS Export Suite**
GeoTIFF with Moon 2000 CRS, ESRI .tfw world files, and GeoJSON tie-points.

## 8. Space Mission Applications & Impact

MISSION VALUE

Sub-pixel hazard mapping for pinpoint landing site identification and boulder risk mitigation.

High-precision coregistration of IIRS hyperspectral 3-micron water-ice bands with OHRC optical maps.

Real-time 210ms edge compute pipeline enables autonomous descent navigation for future lunar landers.